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Customer Churn Analysis and Prediction

SW and methods for statistical analysis of economic data

20 June, 2025

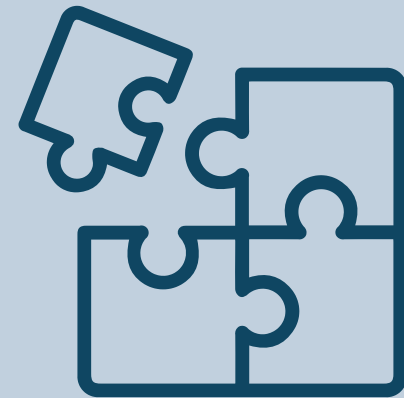


Project Objectives



Predict Customer Churn

- Build a model to identify customers likely to leave the service, and discover key features.



Compare ML Models

- Evaluate logistic regression, SVM, and random forest for performance and interpretability.



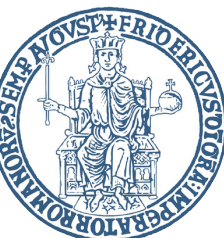
Improve Customer Retention

- Provide actionable insights to help the business reduce churn.



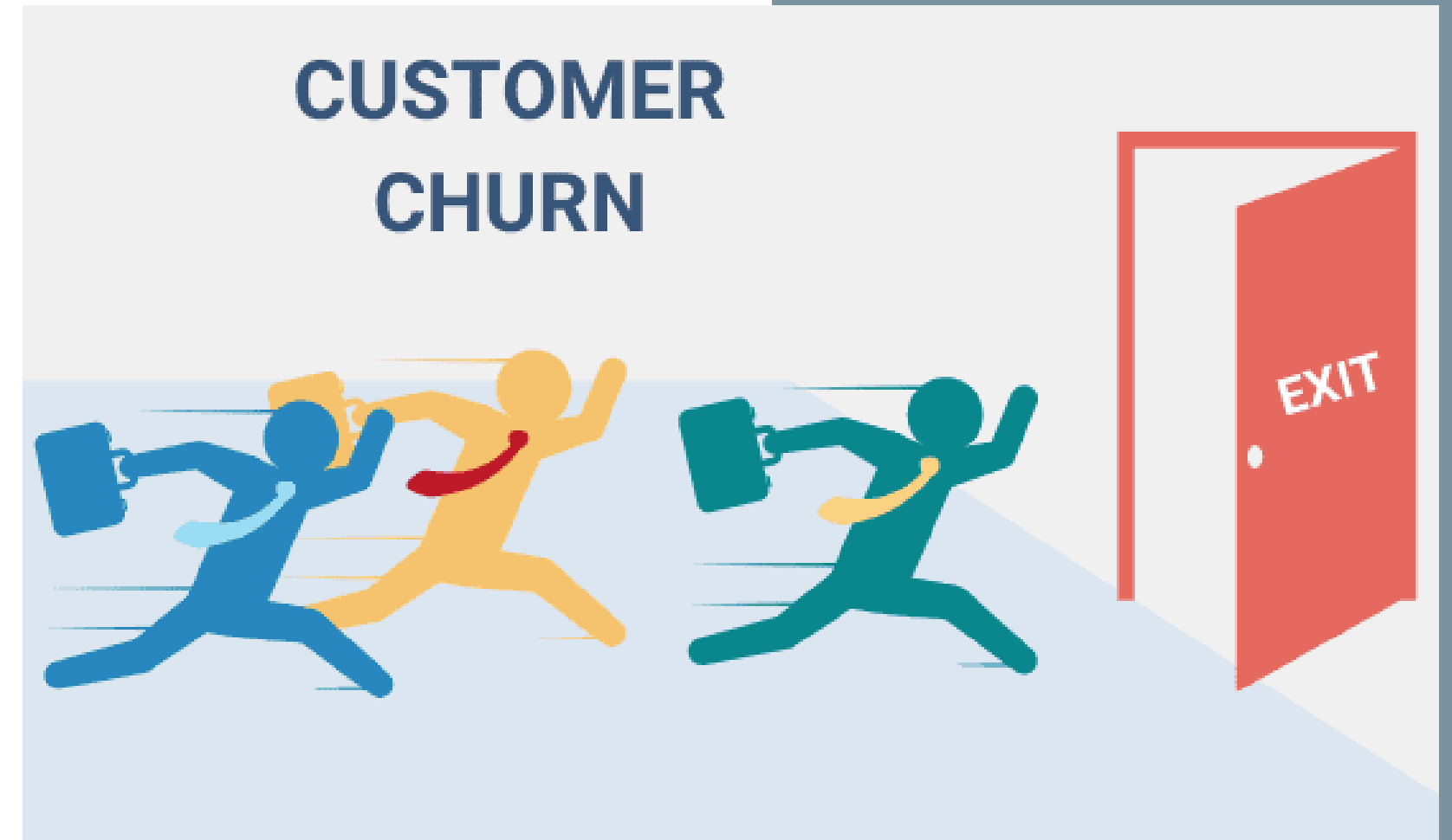
Ensure Explainability

- Use LIME and SHAP to explain both global and individual model decisions.



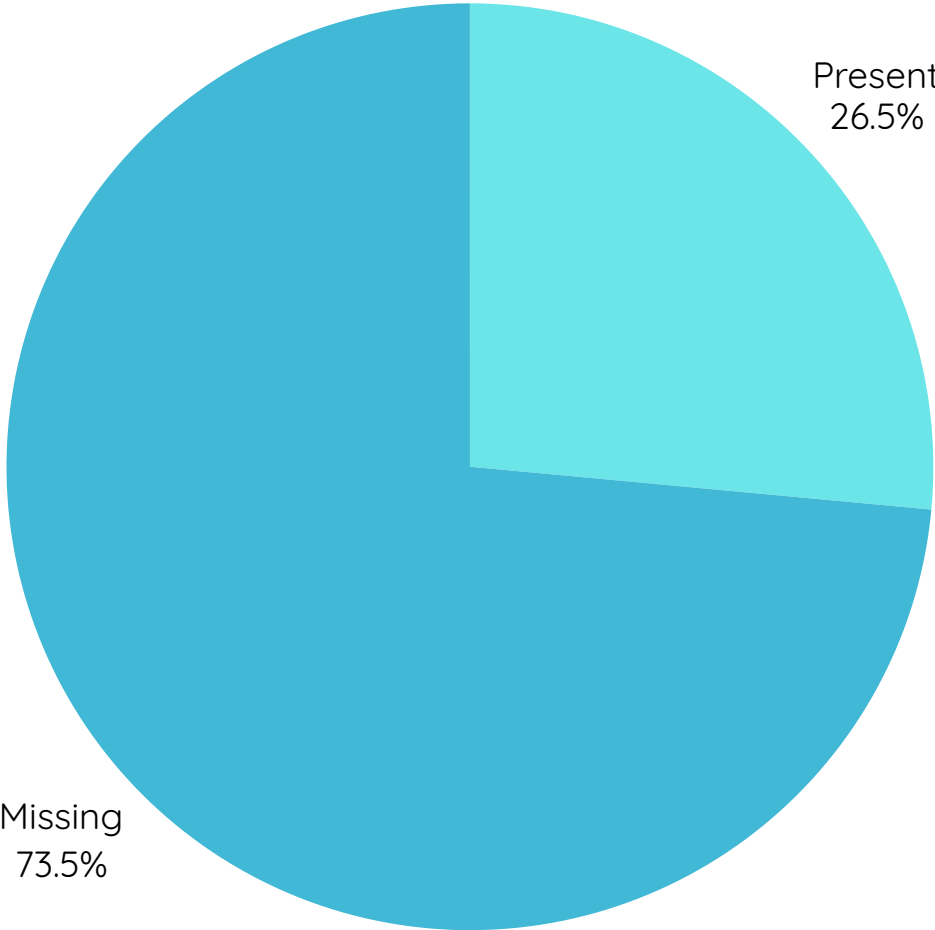
Understanding the Data

- Rows: 6687 customer records
- 16 Categorical & 13 Numerical
- Columns: 29 variables including:
 - Demographics: age, state, gender.
 - Service Info: contract type, internet plan, international plan.
 - Usage Metrics: monthly charge, total charges, avg. GB download.
 - Support Metrics: customer service calls.
 - Churn Details: churn label, churn reason, churn category.



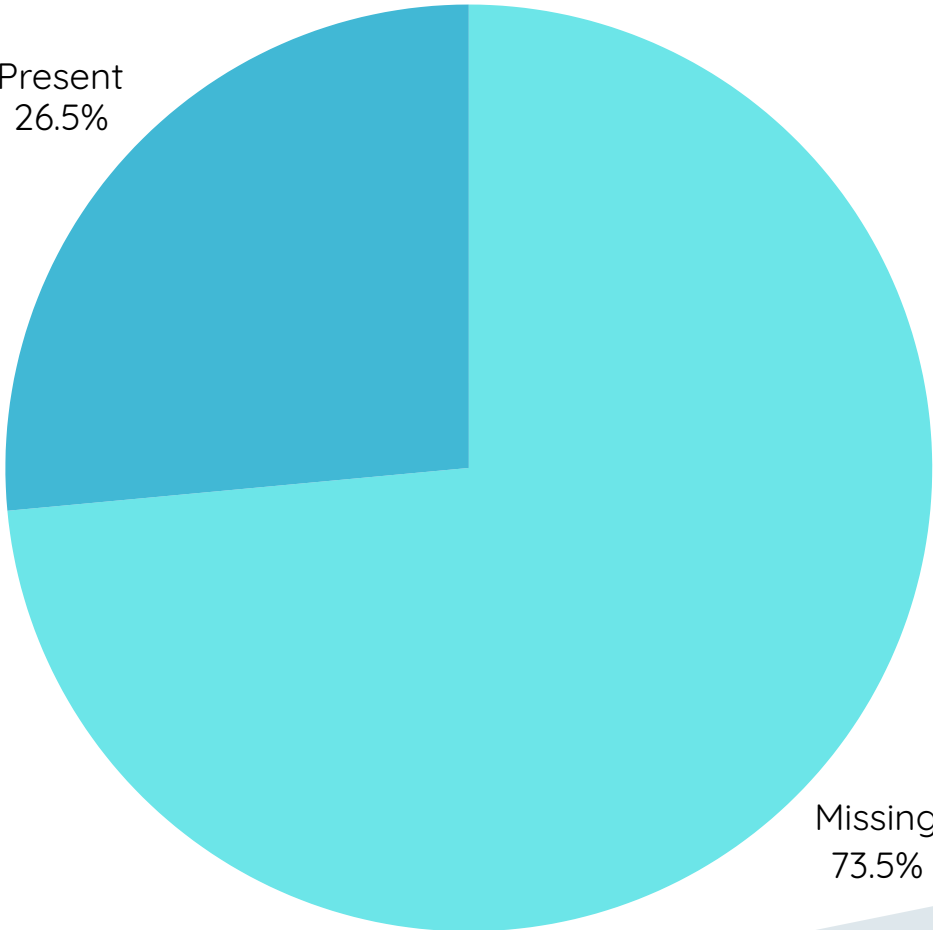
Handling Missing Data

- Missing data wasn't extensive, but some important columns like churn reason had gaps.
- Custom check for "NA", blanks and "".

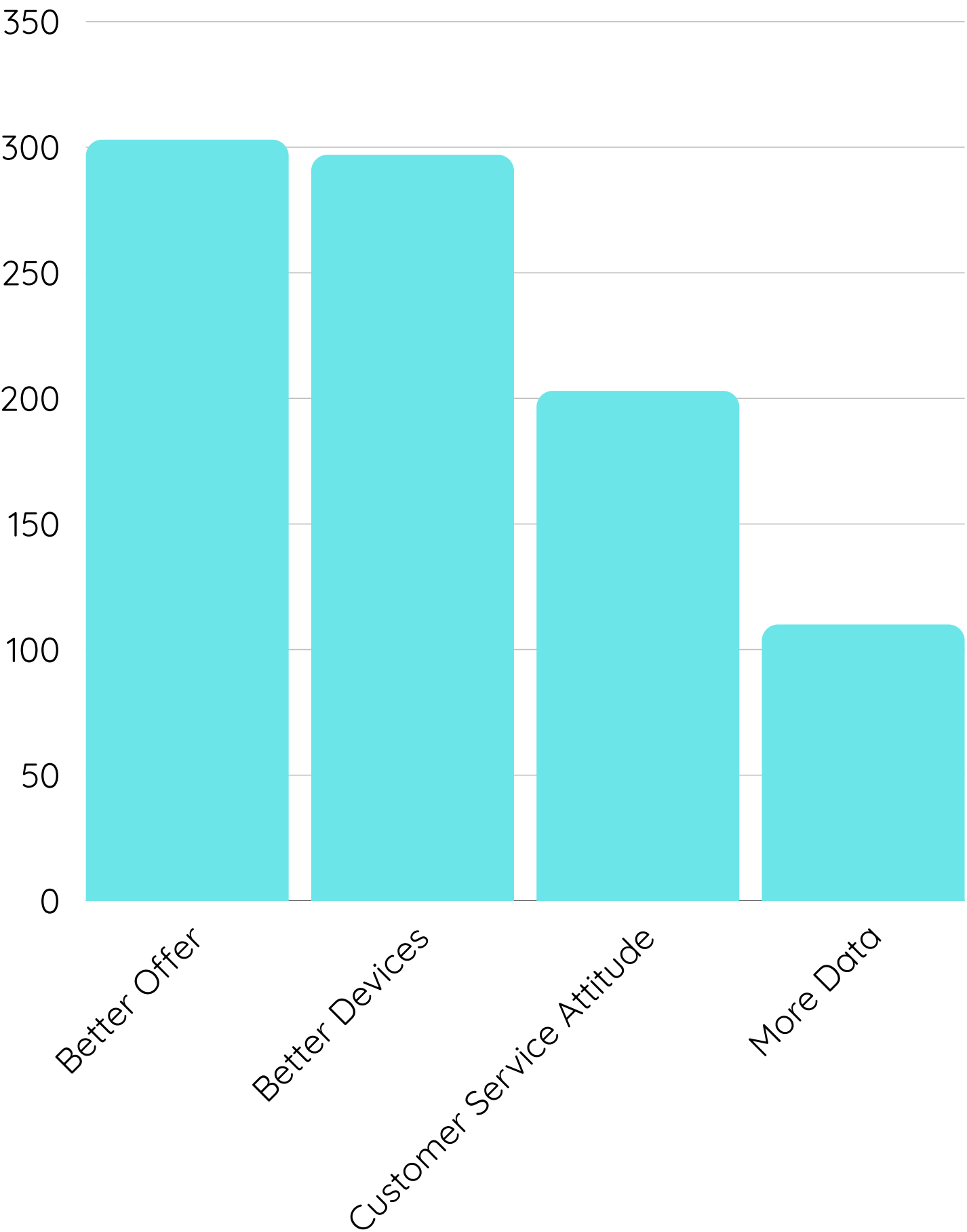


Churn Category

Churn Reason

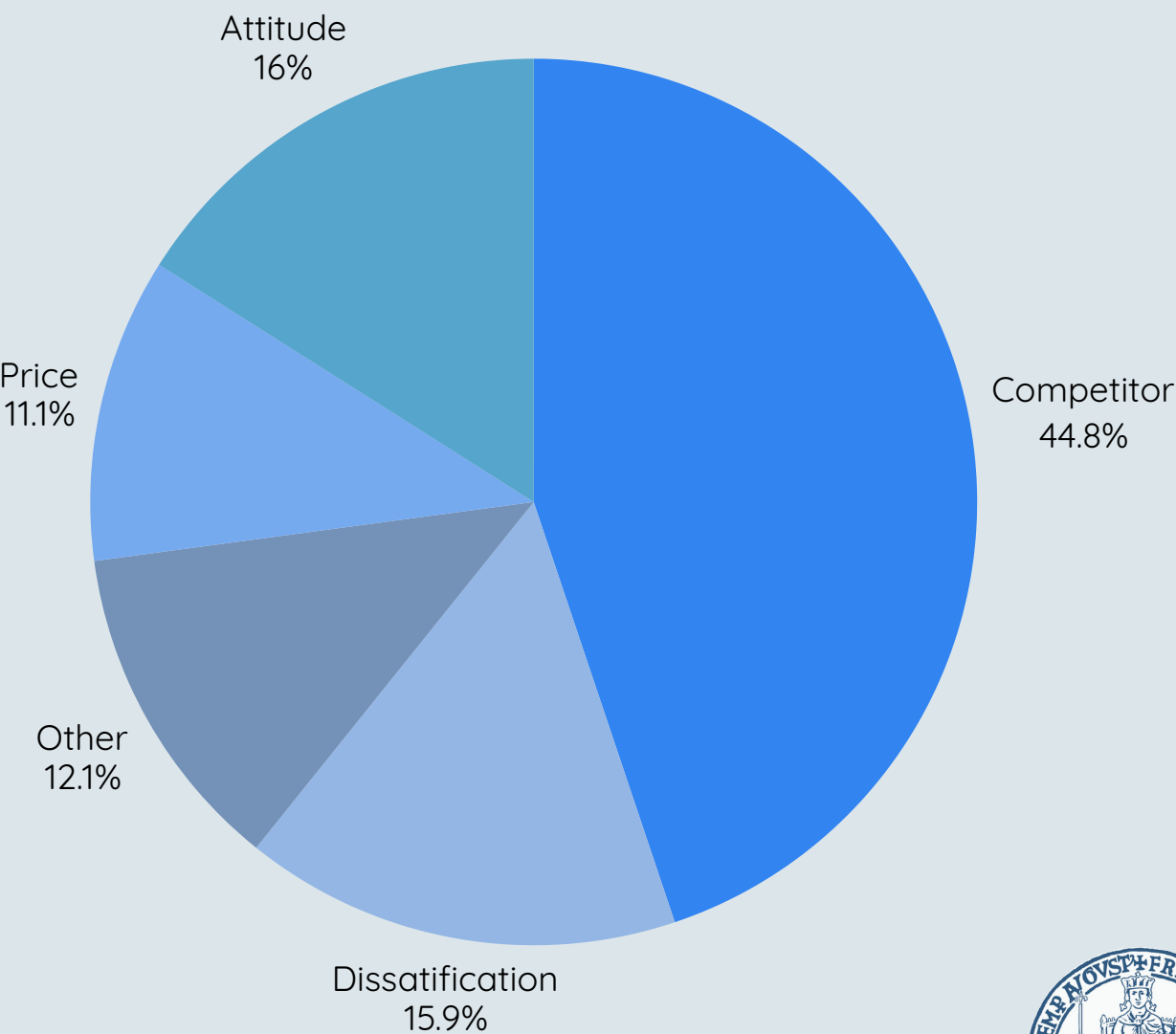


Churn Reason



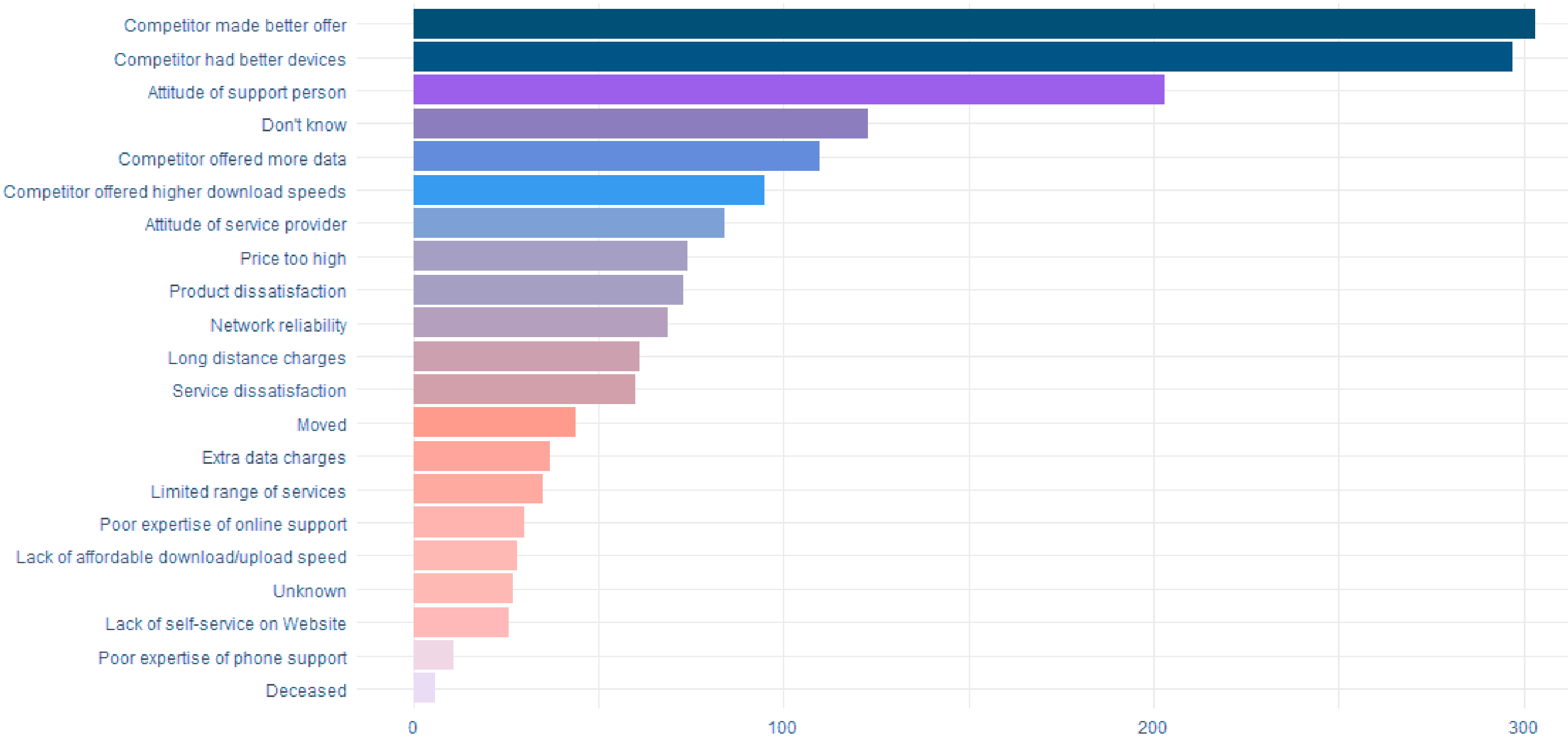
- Total Number of Customers: 6687
- Total Number of Churned Customers: 1796
- Rate of Churned Customers: 26.86 %
- The most cited reason was competitors, which is actionable insight for business

Churn Category

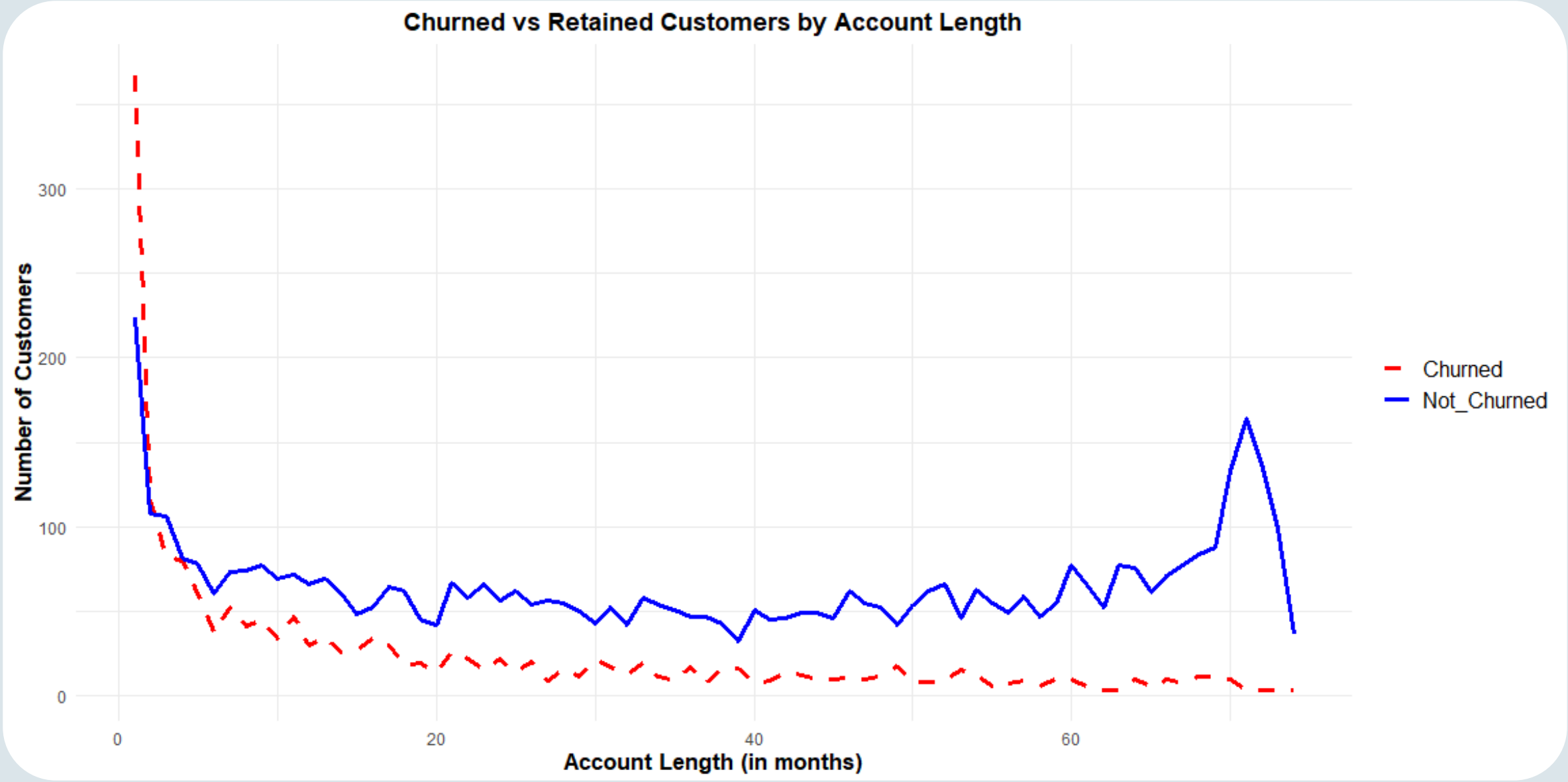
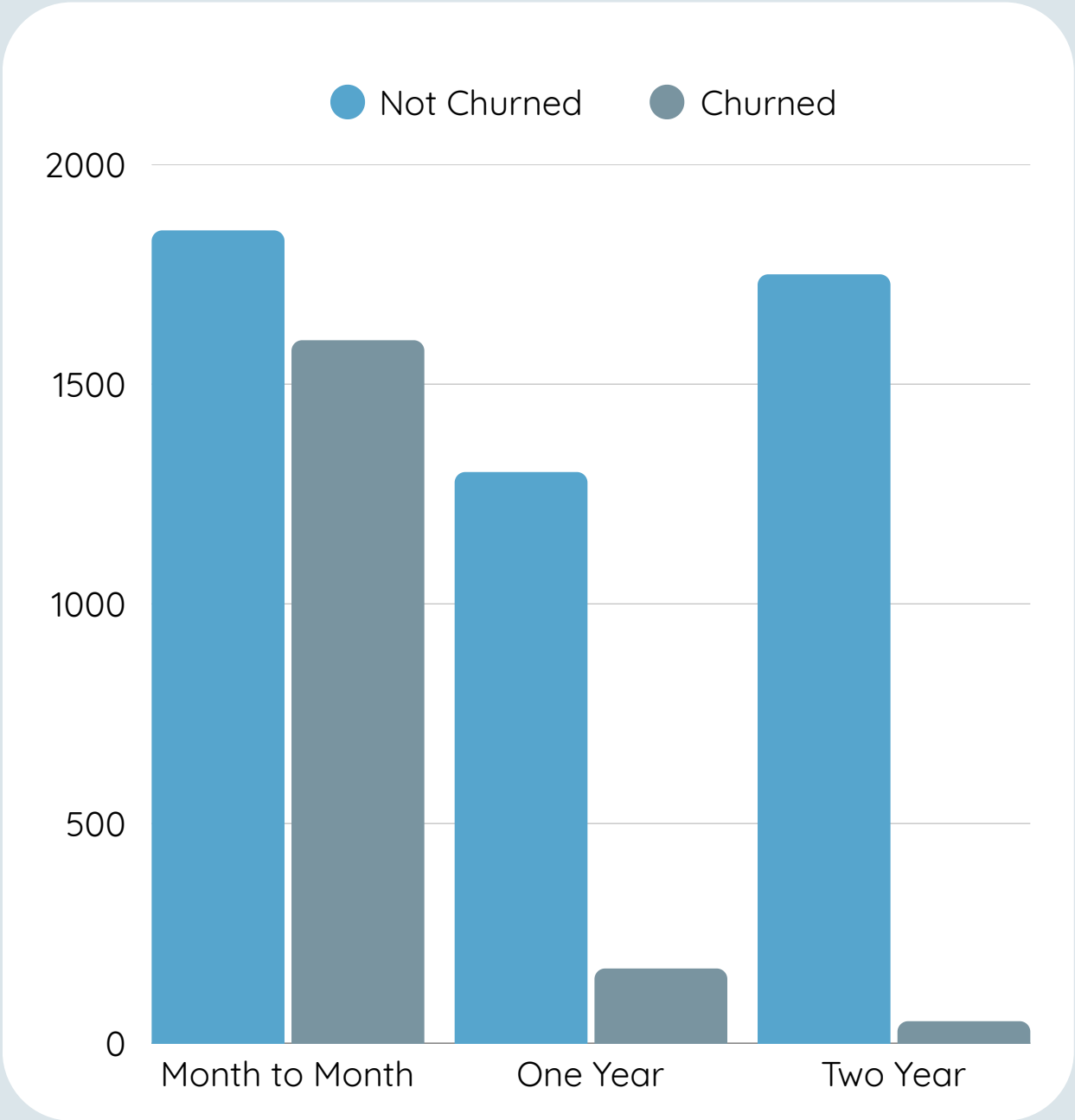


Count of All Churn Reasons

Churn Reason



Contract Type



Feature Engineering

Dropped Irrelevant Columns:

- Removed identifiers: customer id, phone number, etc.
- Dropped high-missing explanation fields not used for prediction: state, etc

Transformed Features:

- One-Hot Encoding:
 - Converted categorical variables (e.g., contract type, intl plan) into numeric format using one-hot encoding.
- Scaling:
 - Standardized numeric features to zero mean and unit variance.

Target Variable

- Converted churn label to a binary factor: Yes vs No



Model Building

Selected Algorithms:

- Logistic Regression
- Support Vector Machine (SVM)
- Random Forest

Why These Models?

- Represent a balance between interpretability and predictive power.
- Allow meaningful comparison of linear vs. non-linear approaches.

Workflow:

- Used tidymodels to streamline model creation, tuning, and evaluation.
- Built separate workflows for each model using the same preprocessing recipe.



Hyperparameter Tuning



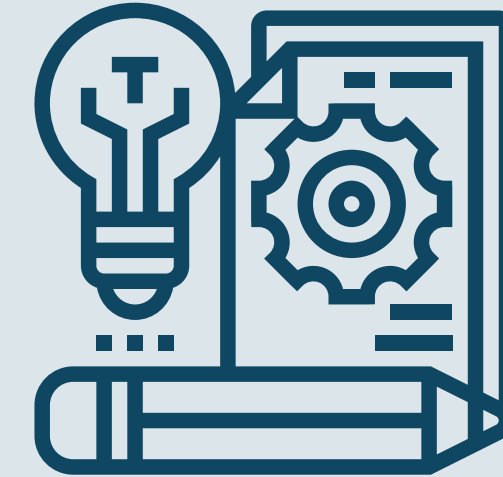
Cross-Validation Strategy

- Used 5-fold cross-validation on training data.
- Stratified sampling to preserve churn rate in each fold.



Tuning Grids

- Logistic Regression: penalty, mixture.
- SVM: cost, rbf_sigma (controls decision boundary flexibility).
- Random Forest: mtry, min_n (feature subset size & min node size).



Implementation Plan

- Evaluation Metric:
- ROC AUC chosen over accuracy due to class imbalance (~27% churn)



Model Comparison

Logistic Regression

penalty <dbl>	mixture <dbl>	.metric <chr>	.estimator <chr>	mean <dbl>
0.0031622777	0.50	roc_auc	binary	0.9360903
0.0031622777	0.25	roc_auc	binary	0.9360173
0.0031622777	0.75	roc_auc	binary	0.9360156
0.0031622777	1.00	roc_auc	binary	0.9359793
0.0000000001	0.75	roc_auc	binary	0.9359089

SVM

cost <dbl>	rbf_sigma <dbl>	.metric <chr>	.estimator <chr>	mean <dbl>
32.00000000	0.003162278	roc_auc	binary	0.9404423
2.37841423	0.003162278	roc_auc	binary	0.9375430
0.17677670	0.003162278	roc_auc	binary	0.9342147
32.00000000	0.000010000	roc_auc	binary	0.9333507
0.01313901	0.003162278	roc_auc	binary	0.9326139

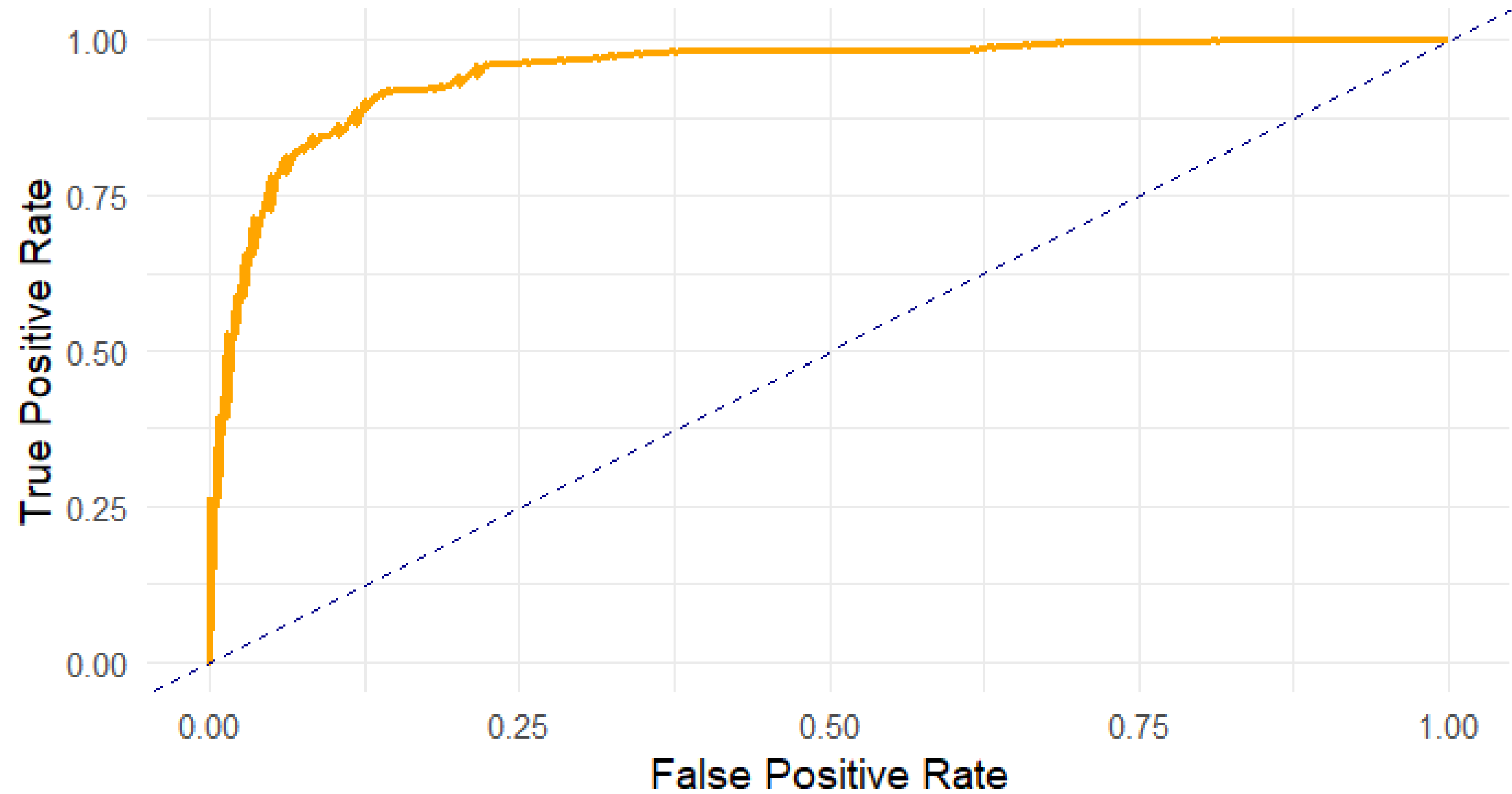
Random Forest

mtry <int>	min_n <int>	.metric <chr>	.estimator <chr>	mean <dbl>
4	10	roc_auc	binary	0.9450872
12	24	roc_auc	binary	0.9433979
25	32	roc_auc	binary	0.9387533
30	38	roc_auc	binary	0.9379208
2	18	roc_auc	binary	0.9377107

Accuracy	0.9067
Precision	0.9150
Recall	0.9617



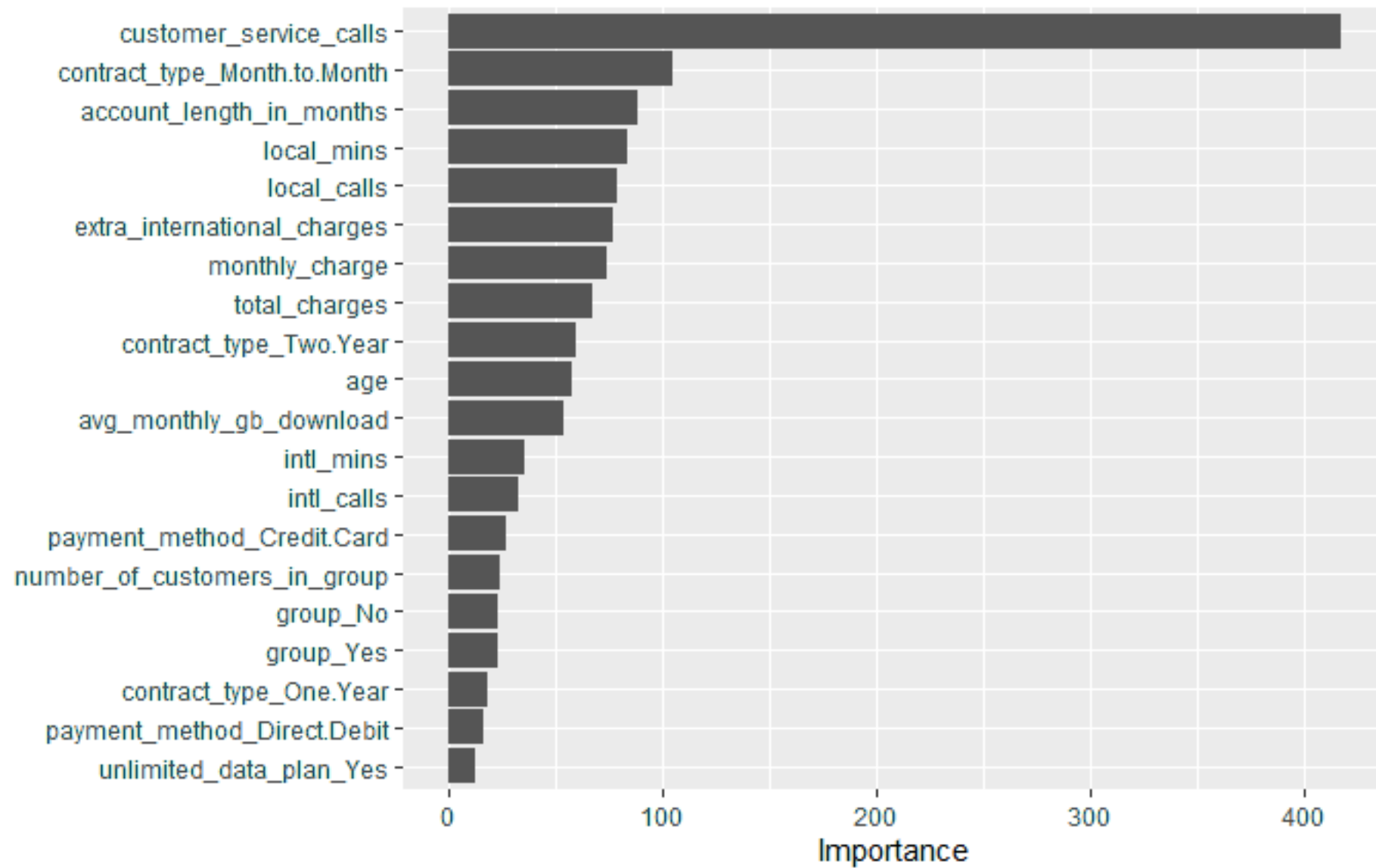
Receiver Operating Characteristic (ROC)



AUC = 0.95



Global Model Explanations

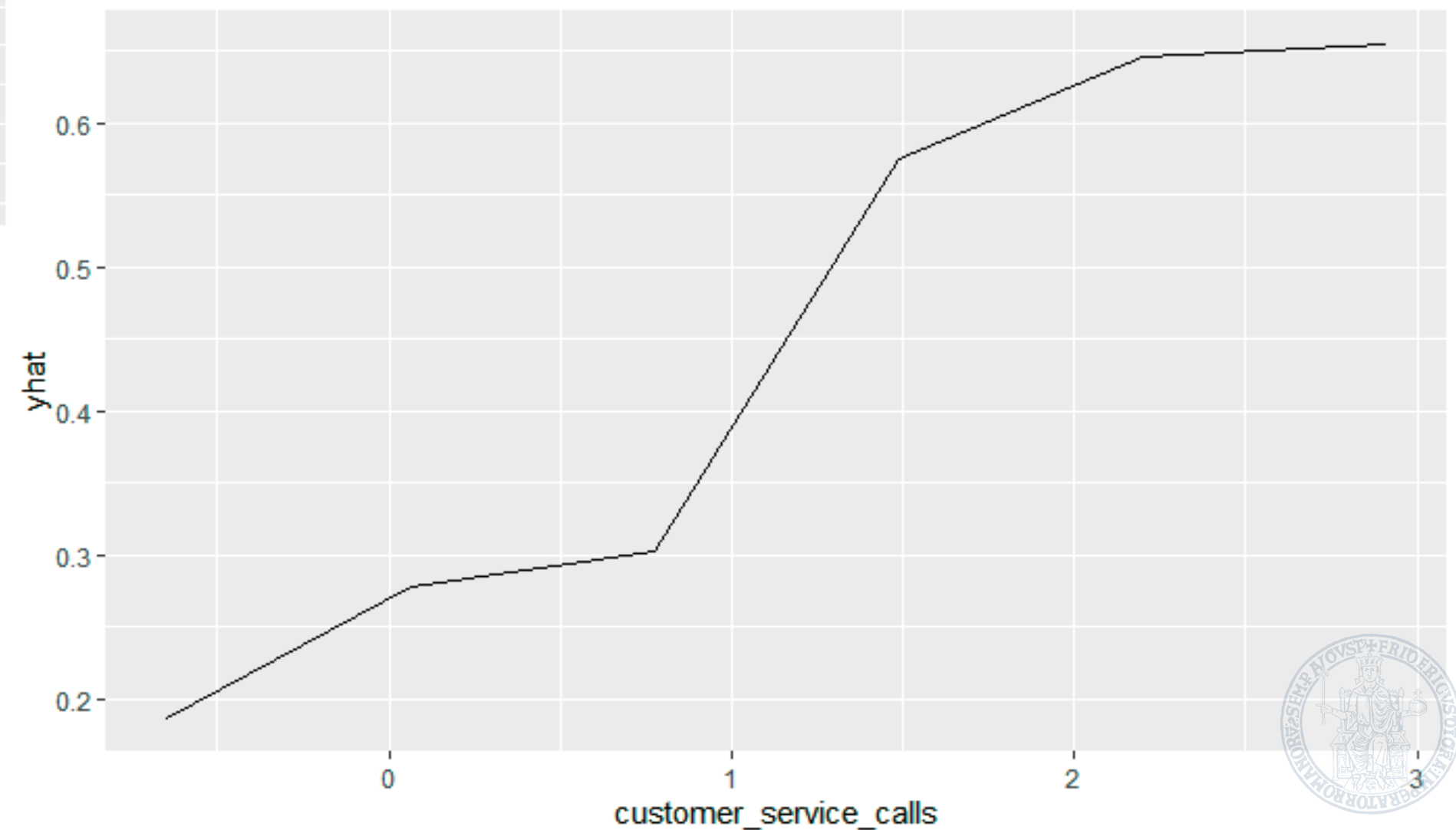


Variable Importance Plot (VIP)

- Ranked features by their contribution to churn prediction.

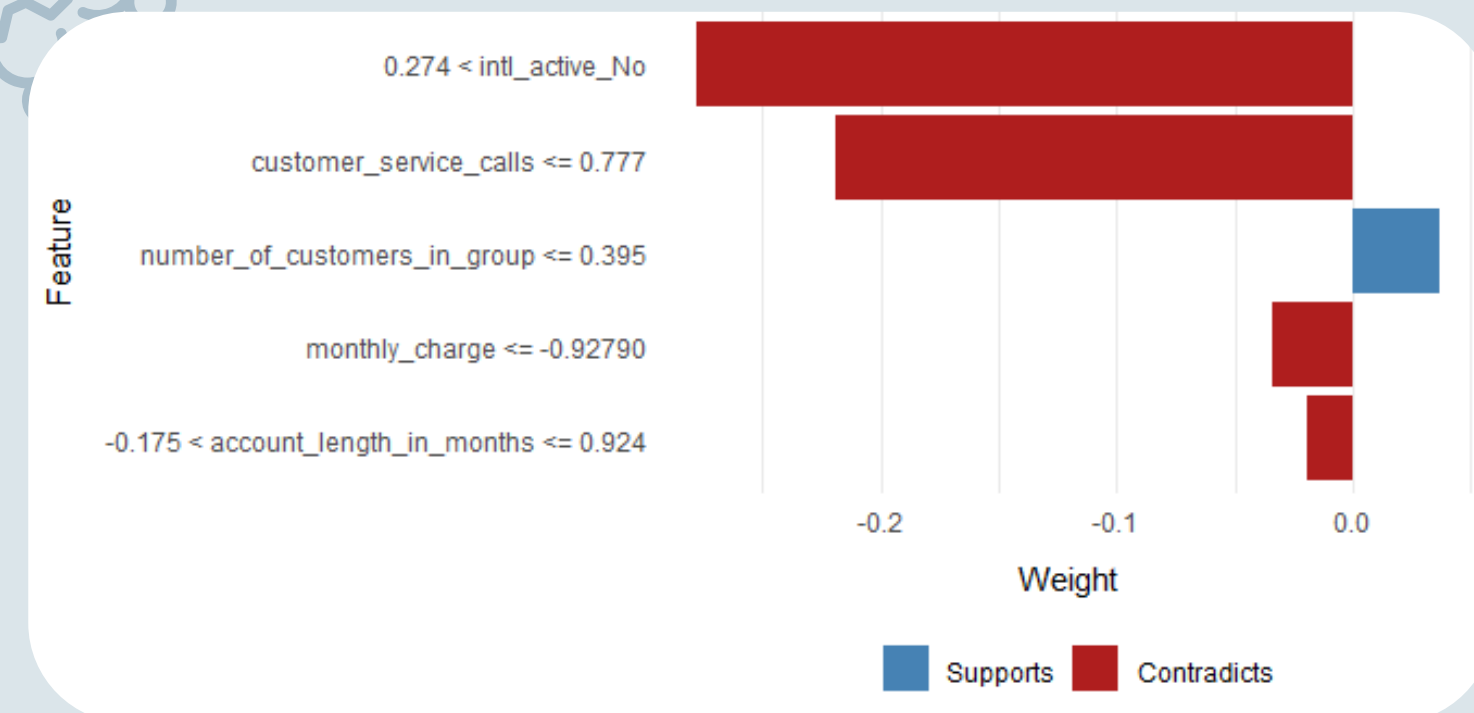
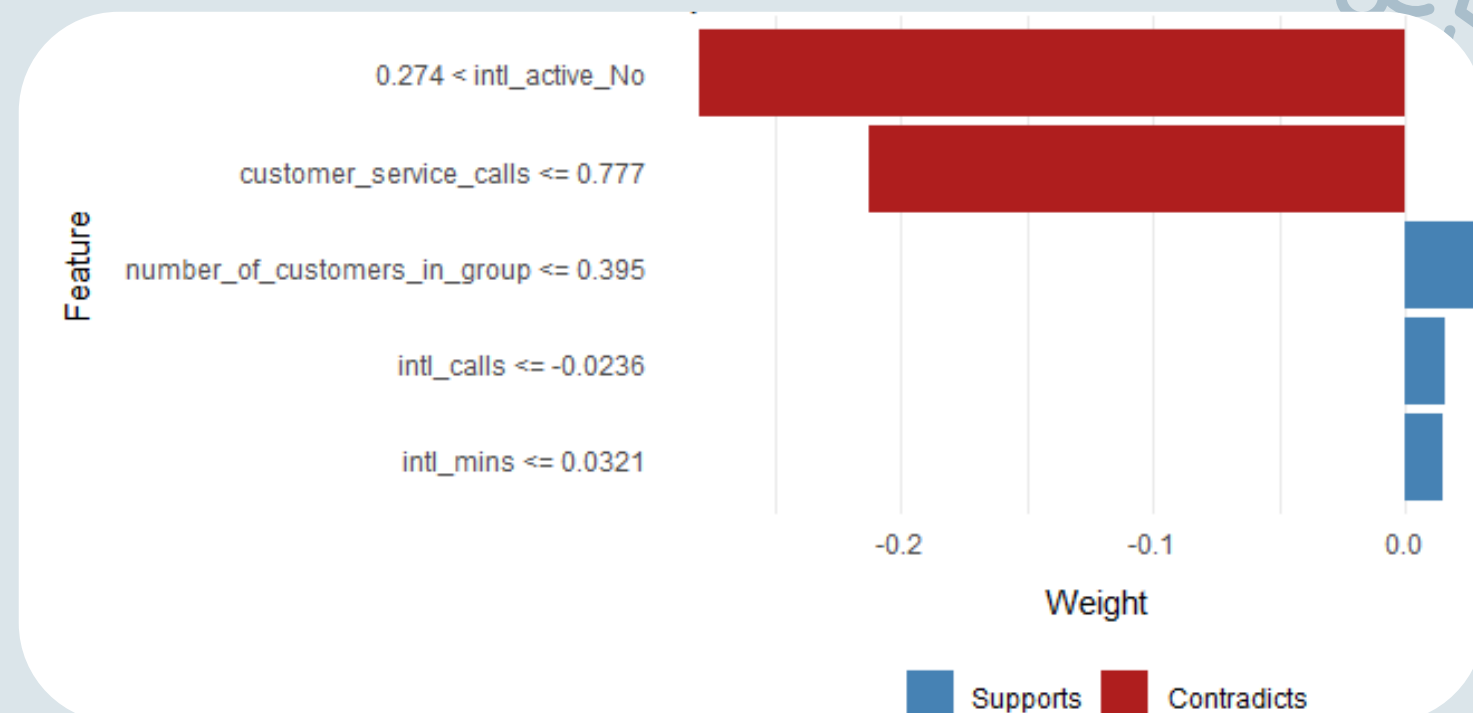
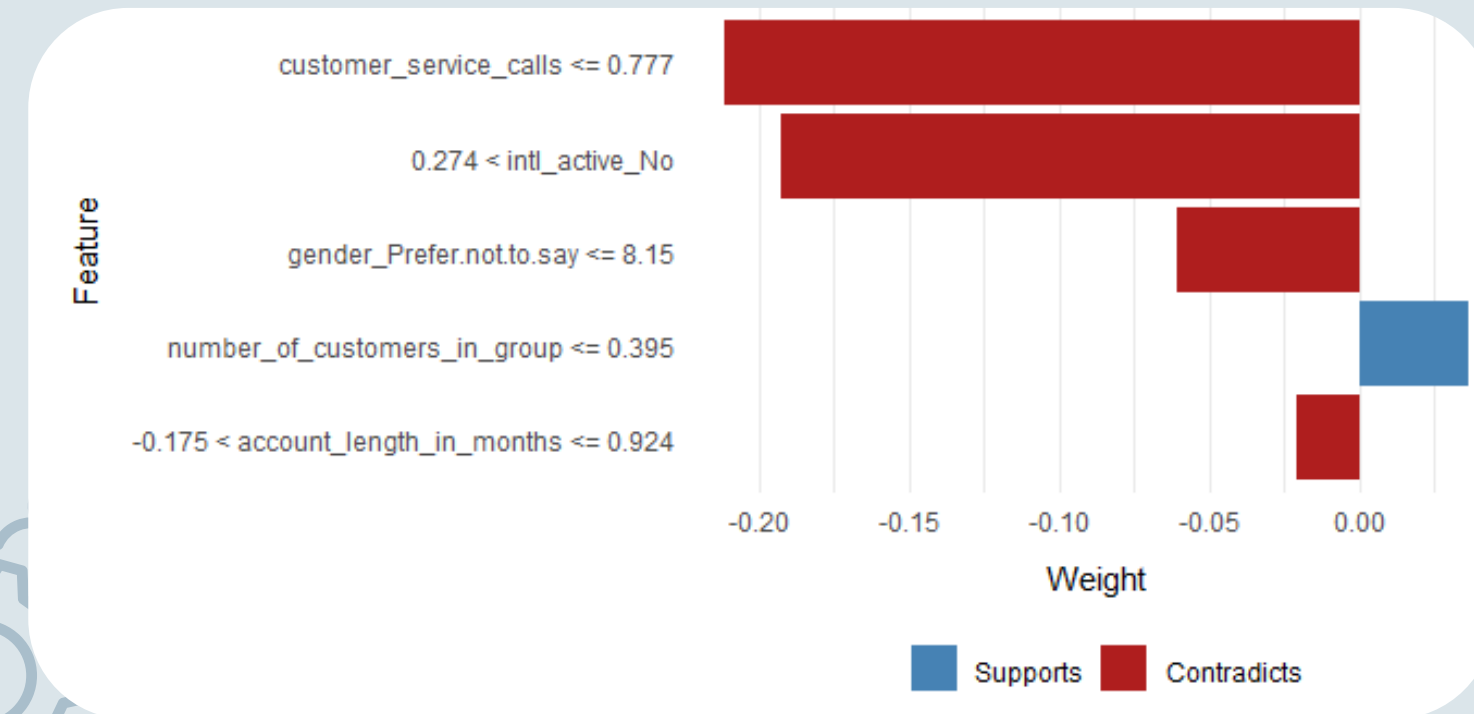
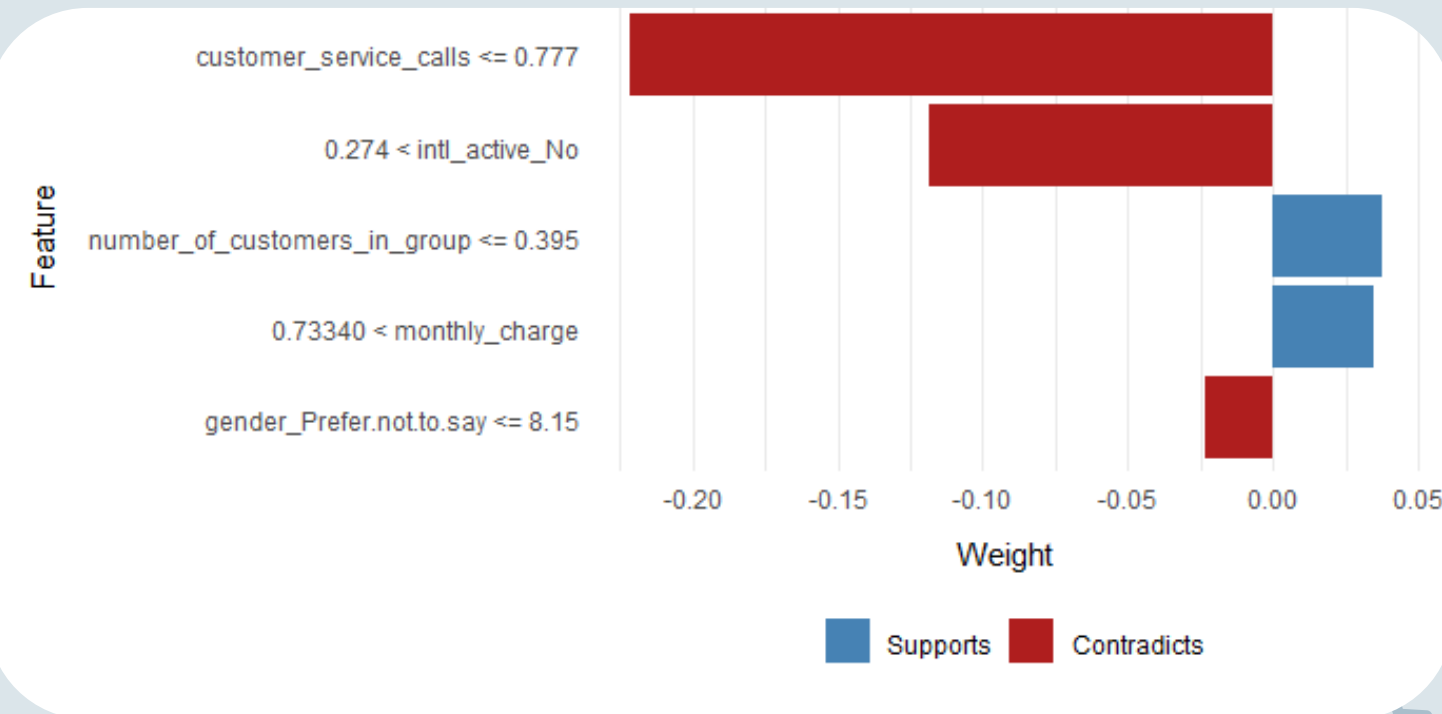
Partial Dependence Plot (PDP)

- Showing the effect of a single feature prediction.



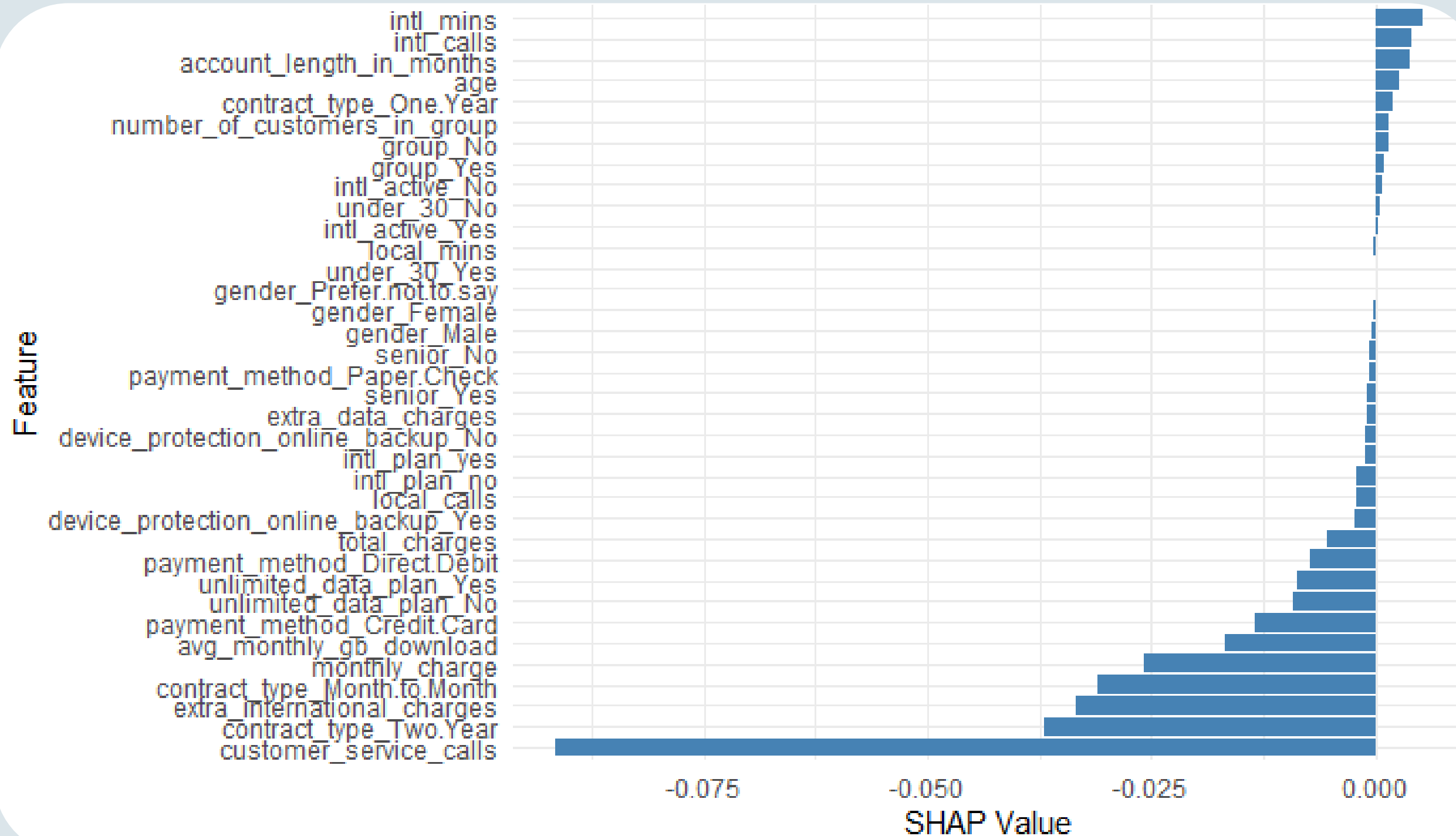
Local Explanations with LIME

- LIME generates a simplified model around a single data point to explain why the model made a specific decision, show which features influenced the churn prediction for individual customer



Local Explanations with SHAP

- SHAP is a method that explains individual model predictions by showing how much each feature contributed to the final decision.





Thank you

